

6 Disagreement and Error

Part 1: Prevalence and Explanation of Moral Disagreement

I claim to have refuted, in the last four chapters, four of the five theories in metaethics: non-cognitivism, subjectivism, naturalism, and nihilism. As is usual in philosophy, the 'refutations' are provisional: those theories have been shown to face grave problems that justify rejecting them as long as any plausible alternative exists. Ethical intuitionism is the remaining alternative. But the received view in the field has long been that intuitionism is a hopelessly naive idea that can be dismissed in a few sentences. 'I propose to ignore this theory', writes Brandt of a related thesis; but most authors simply ignore it with no prior announcement. Ethics textbooks now contain one-paragraph 'refutations' of the doctrine, if they discuss it at all. All of this evidences the esteem in which intuitionism is held these days.¹ So, the naive onlooker might conclude, contemporary philosophers must know some obvious and decisive objections to the theory. What are they?

Most experts take the existence of moral disagreement and error as the main reason for rejecting intuitionism. There are three versions of the argument. One is that the occurrence of disagreement entails that morality is not objective. Another is that intuitionists cannot plausibly explain disagreement or error. The last is that intuitionists have no method of resolving disagreements.

Before elaborating these arguments, let us review the phenomenon their proponents have in mind.

6.1 The prevalence of moral disagreement

There are three main kinds of moral disagreement:

First, disagreements among cultures. Members of a tribe in New Guinea used to eat the brains of dead kinswomen, as part of their mourning ceremonies. Aztec priests used to perform human sacrifices, cutting out the still-beating hearts of victims to feed them to the sun god. Many societies, including our own before the last century, have routinely practiced slavery. All of these practices horrify modern Western observers, yet other societies have found them acceptable, or even obligatory.²

Second, popular disagreements within our society. Some believe that abortion is murder and is therefore among the worst crimes one can commit; others believe it is not wrong at all. Some believe that justice demands capital punishment for heinous criminals; others consider capital punishment uncivilized and unjust. Some think that affirmative action is needed to make reparations for past injustices; others think it adds to the injustice.

Third, theoretical disagreements of philosophers. Some philosophers think the right action is always the one with the best consequences; others say that the ends don't justify the means, and that some kinds of action are wrong even if they produce the best consequences. Some philosophers maintain that the sole good in life is pleasure; others think there are many goods. Some philosophers hold that we ought to value others' interests just as much as our own; others say we are beholden to none but ourselves.

These are a few of the many examples that could be given. They paint a picture of strong, fundamental disagreements as the norm in moral matters. But this is misleading for two reasons. First, many of these disagreements are not fundamentally moral disagreements, but instead turn on differences in the (non-moral) circumstances people find themselves in, or believe themselves to be in. To borrow one of James Rachels' examples, it initially seems that the Eskimos, in their acceptance of infanticide, have very different values from us. But their acceptance of infanticide is explained by the harsh conditions under which they live, in which food is in short supply. In these conditions, families may be unable to provide for all infants, and attempting to do so may threaten the survival of others. Thus, in their circumstances, infanticide makes sense; it need not indicate a lesser respect for life on the part of Eskimos, nor fundamentally different values from ours.³ To take another example, the practice of human sacrifice had roots in religious beliefs. Primitive peoples thought the gods demanded sacrifice. The Aztecs believed the sun would literally stop shining if human sacrifices were not performed; furthermore, they believed the souls of sacrificial victims went to the best afterlife, so the priests were actually doing them a favor. Our society rejects human sacrifice because we reject those beliefs, not because we have fundamentally different values.

Many moral disputes within our society have non-moral roots. Pro-life and pro-choice activists may differ over such questions as whether fetuses are sentient, whether they have souls, and whether God forbids abortion. Disagreements about affirmative action may involve disagreements about the causes of economic inequalities among races. I do not say this accounts for all moral disagreements, but it reduces the prevalence and importance of the disagreements that can be regarded as fundamentally moral.

A second reason why the examples of disagreement are misleading is that they are deliberately selected from a much greater number of moral issues about which there is generally agreement. The cases of abortion, capital punishment, and affirmative action have not been randomly selected from all moral issues; they are selected for attention because they are among the most notorious subjects of controversy. The proponent of the argument from disagreement does not draw our attention to *paradigmatic* instances of murder—say, killing a store owner during an armed robbery so that one cannot later be identified. Nor does he raise the issue of the wrongness of indiscriminate lying, stealing, and breaking of promises. Those things are closer to the core of our moral code, but they do not attract attention or become subjects of public discussion. Public discourse naturally focuses instead on areas of disagreement. This gives us an exaggerated impression of the importance of such disagreement.

The disagreements of philosophers too are somewhat exaggerated. Philosophers agree on paradigm cases of wrongful actions—killing people for fun, torture, typical cases of theft. Moral philosophers argue with one another largely by appealing to hypothetical cases about which we generally have the same intuitions. Indeed, if we did not agree on most cases, we could have no interesting discussion—we might not even understand one another's words, since if your conception of the good bears no resemblance to mine, then I will not understand your words as referring to goodness at all, nor can I credit you with any rational arguments on the subject. What philosophers disagree about is what *account* is to be given of the paradigm cases. For example, they may differ over whether murder is wrong because it violates a right to life, or because it decreases the total amount of value in the world.

None of this is to deny the existence of important disagreements about values. It is only to say that these disagreements are less prevalent than is sometimes thought, and that they take place against a backdrop of substantial moral agreement.

6.2 The idiot's veto

The first version of the argument from disagreement claims that the existence of disagreement about morality is incompatible with morality's being 'objective'.⁴ I shall not dwell on this argument long, since few philosophers would be guilty of advancing it. The argument's premise is that a claim is 'objectively' true only if everyone agrees or would agree with it (perhaps after hearing the arguments for it).

This premise introduces what I call an 'idiot's veto'. The Idiot's Veto is the principle that one can bar any (would-be) fact from the realm of objectivity merely by failing to apprehend it. In effect, the premise grants individuals with limited cognitive abilities or stubbornly skeptical dispositions a *veto power* over any would-be objective truth. Thus, suppose a physics teacher encounters a student who refuses to accept the Second Law of Thermodynamics (such individuals are the source of the perennial efforts to design 'perpetual motion machines'). According to the Idiot's Veto, this would rule out the objectivity of the Second Law of Thermodynamics.

No interesting sense of 'objective' makes that true. Take the definition according to which objectivity is a matter of independence from observers. On this interpretation, the Idiot's Veto claims that for a thing to exist independently of observers, everyone must agree that it exists. This is a *non sequitur*. The definition of objectivity says nothing about agreement.

A second conception takes an 'objective' truth to be one that has been or can be *proven*. On this interpretation, the Idiot's Veto maintains that if at least one person would not agree with $\$x\$$, then $\$x\$$ has not been proven. But on any ordinary understanding of 'proof', it is possible for a person to not accept the conclusion of a proof—whether through lack of understanding, mistake, bias, or what have you—and their failure to believe that conclusion does not invalidate the proof. A proof is, roughly, an argument with no flaws; someone's mere failure to accept the conclusion of an argument does not constitute a flaw in the argument.

Of course, one could simply *stipulate* that 'objective' mean 'would be agreed to by everyone'. But this would deprive the argument of any philosophical interest. Not only would this be a misleading use of the term, but it would be a use according to which there may well be no proposition whatever that is 'objective'. The interest of the debate about moral

objectivity comes from the anti-realists' view that ethics differs in some crucial way from other, more 'objective' fields such as physics and mathematics. The Idiot's Veto identifies no such difference. It implies that the Second Law of Thermodynamics is not objective, as we have seen. It implies that the Theory of Evolution is not objectively true, since Creationists reject it. And it implies that the very existence of the material world is not objective, since Bishop Berkeley rejected it.

To give this version of the Argument from Disagreement any plausibility, we must qualify its premise. Perhaps the premise should be that if some *fully rational, fully informed* person would not agree with a claim, then the claim lacks objectivity. But rational and informed people still make mistakes. One who makes a miscalculation, or who misunderstands or fails to understand a concept, is not thereby irrational, misinformed, or uninformed. Presumably, someone's making an error about x does not refute x 's objectivity in any interesting sense, so the premise needs to be qualified again. We may also add a qualification concerning bias, since failure to agree with x due to bias should not impugn x 's objectivity. Our new version of the premise (no longer an Idiot's Veto) is this: If some fully rational, informed, and unbiased person, who correctly and fully understands x and makes no mistakes, would not agree with x , then x is not objectively true.

Perhaps the premise in this latest version is correct. But now the problem is that of showing that there are any moral disagreements of the relevant kind. While we can readily observe disagreements about moral issues, few if any humans are completely rational, informed, or unbiased. And how shall we establish that both parties to a disagreement correctly and fully understand the issue and have made no mistakes? There are many ways in which humans can make mistakes, most of which are common in moral matters, as we see in the next section.

6.3 Can intuitionists explain disagreement?

The second version of the Argument from Disagreement asserts that the evidence of moral disagreement strongly favors anti-realism over intuitionism, because anti-realists are better able to explain disagreement than intuitionists are. Anti-realists understand inter-cultural disagreements as being due to differing forms of life, together with the absence of any facts determining the 'correct' form of life. They understand interpersonal disagreements as arising from differing personal preferences, together with the absence of any facts determining the 'correct' preferences. The intuitionist, in contrast, must say people are subject to widespread error, but he has no account of why this would be so. In short:

1. There is widespread disagreement about values.
2. If intuitionism were true, there (probably) would not be widespread disagreement about values.
3. Therefore (probably), intuitionism is false.⁶

6.3.1 The caricature

How is intuitionism supposed to preclude the possibility of widespread error? Here is one popular answer; I call it 'the Caricature':

Caricature

Intuitionists hold: (i) All moral questions can be resolved by *direct* appeal to intuition. There is no place for reasoning or any other process of investigation in moral philosophy. (ii) Intuition is infallible. And (iii) everyone knows these things. It follows that everyone can immediately obtain an answer to any moral question by simply consulting his intuition. These answers must all agree, since intuition is infallible. And since everyone knows these facts, no one will ever form a moral belief by any other means. So everyone will agree.⁷

If the history of philosophy has seen a less charitable misreading of a major philosophical position, I do not know what it is. I know of no one who has ever held the views there attributed to intuitionism. Intuitionists have repeatedly denied the claim of infallibility in particular.⁸ Nowell-Smith supports the attribution with the following quotation from the eighteenth-century intuitionist Samuel Clarke:

*These things are so notoriously plain and self-evident, that nothing but the extremist stupidity of mind, corruption of manners or perversity of spirit can possibly make any man entertain the least doubt concerning them.*⁹

Nowell-Smith produces this quotation while discussing the intuitionist's alleged inability to accommodate moral disagreement; he neglects to inform his readers what 'these things' refers to in the passage, leaving them perhaps to infer

that it refers to all moral truths. In fact, Clarke is referring to a specific series of moral principles that he has listed, principles such as the following:

[I]n Men's dealing and conversing one with another, 'tis undeniably more Fit . . . that all Men should endeavour to promote universal good and welfare of All, than that Men should be continually contriving the ruin and destruction of All.⁹

While there are important moral matters that people disagree over, the above principle is not among them. The only of Clarke's examples that are at all controversial are those concerning obligations of and to God, where the source of controversy is metaphysical, rather than ethical. Thus, the quoted passage does not justify attributing to Clarke the view that we are infallible about all moral matters—to say nothing of other intuitionists.

In any event, if anyone has held the absurdly strong and simplistic views described in the Caricature, I do not.

6.3.2 The prevalence of non-moral disagreement

Perhaps I have been unkind to the argument from disagreement. A more subtle version holds, not that intuitionism rules out any moral disagreement, but that intuitionism *renders improbable* disagreements of the kind and number that we find in moral philosophy. Disagreements in moral philosophy are very widespread, are often apparently unresolvable, and sometimes concern fundamental questions of the sort that we would expect intuition to speak to. It is this specific pattern of disagreement that the intuitionist allegedly cannot explain. Though they reject infallibility, presumably intuitionists at least believe that intuition is generally reliable at producing correct beliefs; otherwise, it could hardly account for moral knowledge. But the observed pattern of moral disagreement impugns the reliability of moral intuition and moral reasoning, since if we had available a reliable method of identifying moral truths, there should be relatively few disagreements, and disagreements should frequently be resolved by application of that method.¹¹

I disagree. I don't think the availability of moral knowledge is incompatible with frequent and persistent moral disagreements. This subsection brings out the first part of my case for this: the prevalence of non-moral disagreements with the same problematic features as moral disagreements. Consider some examples:

1. Who shot JFK? Forty years after the assassination, many remain convinced that there was a conspiracy, possibly involving Cubans, the mafia, or the CIA. Others are equally certain that a lone gunman, Lee Harvey Oswald, was responsible.
2. What is the best football team in America?
3. How should quantum mechanics be interpreted? Einstein and Bohr debated whether the formalism of quantum mechanics was complete, without issue. To this day, there remain multiple interpretations of quantum mechanics, each with their own advocates.
4. What are the actual practices of other cultures? Anthropologists for decades took there to have been many cultures in which cannibalism was common. In 1979, William Arens argued that this was a myth, that in fact no culture ever practiced widespread cannibalism. Anthropologists disagreed sharply in their assessments of Arens' work. Similarly, a heated debate arose when in 1983 Derek Freeman purported to show that Margaret Mead's famous account of life in Samoa was entirely inaccurate. Both controversies aroused strong emotions and continue to this day.[12]
5. What are the economic effects of government social programs? Some think they have been essential in ameliorating the great poverty and suffering that would otherwise be the lot of the lower classes in a capitalist country. Others think those same programs actually harm the poor.[13]
6. What religion, if any, is correct? We've had millennia to think about it, and we haven't even agreed on whether there is a God, let alone what He wants.
7. Are there paranormal phenomena? I suspect most readers will answer no. But a quarter of Americans believe in astrology, half believe in ESP, and a quarter believe they have had a telepathic experience.[14]
8. What causes illness? The same cultures that have very different moral beliefs from ours also accept very different answers to this question. Many think illnesses can be caused magically by others' ill will. Western society accepted for centuries that diseases were caused by imbalances of the body's four humors.

Many more examples could be given. The disagreements in examples 1-7 have the following features:

- i) They are widespread among people who have an opinion about the issues; it isn't just a few people going against a consensus.
- ii) They are apparently unresolvable. Anyone who thinks they can resolve who shot JFK by 'just going in and looking at the facts' is invited to try. You may well figure out 'the answer' (by your lights), but only the uninitiated will imagine they have a chance of convincing the other side(s).
- iii) They concern objective, non-evaluative facts. Indeed, such questions as who shot JFK and whether a particular culture practiced cannibalism are paradigms of objective questions.

In the case of example 8, there was an agreed upon answer in our society for several centuries, but this answer was completely wrong.

I have listed a number of examples to illustrate the scope of human disagreement. People disagree about historical events, about apparently straightforward matters of empirical fact, about things allegedly established by science, about the interpretation of accepted theories, and about many philosophical matters, among other things. Even when we agree, that is no guarantee of our being right, as example 8 shows. I have, admittedly, given a partial view of our intellectual life: I have intentionally selected notable cases of strong and persistent disagreements. But then, the anti-realist has done likewise for the moral realm in calling forth such examples as abortion and capital punishment.

Persistent non-moral disagreements are common. Yet virtually no one takes this to refute the possibility of knowledge in such fields as history, anthropology, and economics. I suspect few would draw even the slightly skeptical conclusion that we cannot gain knowledge concerning the specific issues about which we disagree (even if we haven't yet done so). How, then, is moral disagreement supposed to refute the possibility of moral knowledge?

Perhaps there is something especially problematic about disagreements in areas where knowledge depends on intuitions, as opposed to areas where knowledge depends on other means, such as observation and scientific reasoning. If there is, I have no idea what it is. Intuitionists need not and do not maintain that knowledge based on intuition is superior to empirical knowledge; they need not take reasoning based on moral intuitions to be more reliable than science, for example. Opponents of intuitionism are simply applying an epistemological double standard: If scientific and other empirical reasoning fails to produce agreement in many cases, yet this does not invalidate it as a source of knowledge, then why should intuition's similar failure to produce agreement invalidate it as a source of knowledge?

Perhaps the problem is that in the case of moral philosophy, we have many disputes all within the same subject matter, whereas the non-moral disputes are isolated instances distributed across many different subjects. But this is not so. There are numerous disputes on the subject of religion, even its non-moral claims; all branches of philosophy; sports, particularly the merits of various players and teams and what people 'should' do (in a non-moral sense) at various points in a game; the relative abilities of specific individuals one may know (for example, is Sally smarter than me?); and everything bearing on politics, again including non-moral questions. Yet none of these are areas in which objective facts don't exist, nor in which we lack cognitive faculties capable of apprehending facts of that kind.

6.3.3 A menagerie of error

What is surprising about disagreement, such that it calls for special explanation? Beliefs, we think, aim at the truth. If people attain the truth, then they will agree with each other. So widespread disagreement entails widespread error, even though we are ostensibly trying to avoid error. Why don't we just stop using the unreliable belief-forming methods that lead us into disagreements? Why don't we just sit down, taking our logic and critical thinking textbooks in hand, and figure out who is making the bad arguments?

The preceding questions rest on naive assumptions. While beliefs in some sense aim at the truth, people's beliefs may gratify many other passions besides the rational hankering after the truth. Nor do the canons of logic and critical thinking algorithmically determine what is and isn't a bad argument. People seldom satisfy ideal conditions for reasoning, and even the best of cognitive conditions do not render humans infallible. The causes of our beliefs can be highly complex and largely hidden from us. We often cannot articulate some of the reasons for our beliefs. The intuitionist can no more produce an algorithm for computing the correct moral theory than we can produce a general algorithm for computing correct empirical theories.

Critics have charged that the intuitionist has available no account of moral error other than 'blindness', stupidity, or deliberate mendacity.[15] To better assess this charge, let us review some of the sources of error in general:

1. Bias: People who are biased are moved by emotions and desires, other than the desire to have true beliefs and avoid false ones, to view some claims favorably and others unfavorably. In other words, there are some things we want to believe, and others we want not to believe. Often, we want to believe what is in our interests to believe, or to convince others of; what we would like to be true; what would be in the interests of the social group with which we identify to believe; and what coheres with the self- image we want to maintain. These are just some of the possible biases we may suffer from.¹⁶
2. Miscalculation: While performing a calculation, I forget to carry a '1', resulting in an incorrect answer. If no one points it out to me, I may never realize my error. I may fully understand the mathematical principles involved in the problem; be rational, unbiased, and intelligent; and yet still make a mistake.
3. Confusion: This involves failure to distinguish two superficially similar things. For example, the student who thinks that because we cannot talk about anything without using language, we cannot talk about the language-independent world, is committing a confusion.
4. Misunderstanding and lack of understanding: Concepts and theories can be misunderstood. A student who thinks that chaos theory makes room for free will suffers from a lack of understanding of the concept of chaos and perhaps of free will. The philosopher who finds intuitionism absurd because it makes no room for moral debate suffers from a misunderstanding of the theory.
5. Oversight: People may fail to notice relevant considerations, including some of the implications of their statements. Someone asks me whether I think two solid objects can overlap in space. I answer 'No', based on my immediate impression—overlooking such examples as my hand and my body, or a statue and the mass of clay of which it is composed. Relatedly, people may exercise selective attention, that is, they may attend to only some of the relevant features of a situation, whether because of bias or simply because the situation is too complex to take everything in at once.
6. Hasty judgments: People often make a judgment based on an initial appearance, when longer, more careful thought would have resulted in a different judgment.
7. False or incomplete information: Sometimes people do not know all the facts that are relevant to making a judgment, and sometimes they make mistaken assumptions. For instance, two people arguing about gun control may have received radically different information, one by consulting NRA literature, the other by consulting literature from Handgun Control, Inc. Some of their information may be false, while neither has complete information.
8. Unarticulated assumptions: Imagine two people arguing about whether abortion should be legal. One assumes that they are arguing about whether abortion should ever be allowed, while the other assumes they are arguing about whether it should always be allowed. Or perhaps neither of them really knows which of these questions, if either, they intend to discuss.
9. Stubbornness: People may simply refuse to reconsider old beliefs, even when new, relevant evidence appears.
10. Fallacies: These are illusions of reason—forms of reasoning that may superficially appear cogent but in fact are not; for example, equivocation, affirming the consequent, argument ad hominem. Especially important is misplaced trust in authority: people may believe a thing because others—parents, religious leaders, reporters, or other members of their society—have asserted it. But those others may not have known what they were talking about.
11. Forgetfulness: People may forget or simply not notice their own reasons for holding a belief, and they may forget other relevant evidence. In this case, they are in a poor position to recognize when the belief has been undermined.
12. Intrinsic difficulty of issues: Some issues, such as those that are particularly abstract or complex, are more difficult to grasp clearly and make judgments about than others.
13. Inarticulate evidence: Sometimes we have grounds for belief that we cannot articulate. Suppose a friend has been accused of embezzling money from her company. Because I know her well, I know she would not do such a thing.

But I could not prove this to a third party. This sort of inarticulate evidence can cause disagreement.

14. Mental defects: Last and perhaps least, a few people lack cognitive capacities that most others have, or have cognitive capacities whose functioning is seriously distorted, as in the case of schizophrenics, autists, and psychopaths. Though a paranoid schizophrenic may otherwise appear a competent reasoner, you will have no luck reasoning him out of his delusions.

These sources of error are not all independent of each other; some overlap and some influence others.

I cannot claim completeness for my menagerie of error; perhaps there are further sources of error I have overlooked. Given the multitude of ways human beings can go wrong, why do anti-realists think that moral disagreement must be explained by a shortage of objective facts? All of the above sources of error and disagreement operate in non-moral matters whose objectivity is undisputed; there is no obvious reason why they should not also operate in moral matters.

6.3.4 Disagreement is predictable

Proponents of the argument from disagreement, all too eager in their rejections of intuitionism, have neglected to justify the crucial second premise of their argument—

2. If intuitionism were true, there would not be widespread disagreement about values.

As we have seen, there are many ways human beings can go wrong, and there are many persistent disagreements about objective, in principle knowable facts. In light of this, we have no obvious grounds for predicting that there would not be widespread disputes about values.

But now I want to say something stronger: I think that we have obvious grounds for predicting that there would be widespread disputes about values, even if intuitionism were true. This requires looking at the kinds of cases in which disagreement and error are common, in general. There are at least four kinds of cases to consider.

First: Disagreement and error are common where there are strong and frequent biases. Bias is particularly important, since it increases the frequency and seriousness of most other kinds of errors, such as oversight, misunderstanding, forgetfulness, and so on. It probably plays a major role in the disagreements listed in section 6.3.2. Here are some common sources of bias, all of which may affect moral questions:

Self-interest: It would be in my interests, for example, to believe that the things I want to do are morally acceptable, and that the things I want others to do are obligatory. It would be even more clearly in my interests to convince others of these things.

Wishful thinking: I would like it to be true that the world is just; that the way things actually are, is good; that I and my social group are especially praiseworthy; that our practices are justified.

Group interests: It would be in the interests of my social group to believe that we are superior to other groups, that the things we want to do are justified, and so forth.

Self-image: Sometimes the self-images we want to construct affect what beliefs we adopt. If I wish to portray myself as compassionate and forgiving, I may want to reject capital punishment; but if I prefer to portray myself as tough and uncompromising, then I may want to embrace capital punishment.

Second: Disagreement and error are common in areas where people defer to their cultures' traditions. Cultural traditions are a notoriously unreliable source of information. The cultures with the surprising moral beliefs, which are supposed to make us doubt the reality of moral facts, also have a host of surprising non-moral beliefs—about such things as the causes of disease, the structure of the cosmos, human history, and the workings of the physical world—that we know, scientifically, to be false.

Third: Disagreement and error are common in areas where people defer to religion, another unreliable source of information. In section 3.4, I mentioned some false moral beliefs supported by the Bible. Here are some false factual beliefs also apparently supported by the Bible:

Rabbits chew cud. (Deuteronomy 14:6-7)

Some insects are four-legged. (Leviticus 11:20-22)

The Earth is flat. (Matthew 4:8)

$\pi = 3$. (1 Kings 7:23)

The Earth is stationary and rests on a foundation. (Psalms 104:5)

The Earth is about 6000 years old. (Allegedly supported by various parts of the Bible.¹⁷)

Fourth: Disagreement is common in all areas of philosophy. For centuries, philosophers have disagreed about such things as God, free will, universals, *a priori* knowledge, and the nature of the mind, to name a few. The level of disagreement in ethics is no greater than that found in metaphysics, epistemology, philosophy of mind, or philosophy of language.¹⁸

On all four counts, we should expect moral issues to be especially prone to disagreement. My response to the major premise of the argument from disagreement, then, is this: even if there were objective moral truths and we had available ways of knowing them, widespread, strong, and persistent disagreements could still be expected. No anti-realist assumption is required to predict the phenomenon.